KAIZEN MATHS

Lesson starter

SUVAT and Equations of Motion

Multi-Step Motion

Name _____

S1

A car moving at 18 m/s brakes with constant deceleration 2 m/s² until it stops. Find the braking distance.

S2

A particle starts at 6 m/s and accelerates uniformly at 4 m/s² over a displacement of 176 m. Find its final speed.

S3

A cyclist starts at 9 m/s and accelerates at 4 m/s² for 5 s. The cyclist then continues at the new constant speed for 6 s. Find the total distance travelled.

S4

A particle accelerates uniformly from 9 m/s to 21 m/s in 9 s. Use average velocity to find the displacement.

Show your working

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MEC-8084-L2 · S

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KAIZEN MATHS

Exit ticket

SUVAT and Equations of Motion

Multi-Step Motion

Name _____

EA1

A car moving at 20 m/s brakes with constant deceleration 4 m/s² until it stops. Find the braking distance.

EA2

A particle starts at 7 m/s and accelerates uniformly at 2 m/s² over a displacement of 44 m. Find its final speed.

Show your working

KAIZEN MATHS

Exit ticket

SUVAT and Equations of Motion

Multi-Step Motion

Name _____

EB1

A cyclist starts at 5 m/s and accelerates at 4 m/s² for 4 s. The cyclist then continues at the new constant speed for 8 s. Find the total distance travelled.

Show your working

KAIZEN MATHS

Exit ticket

SUVAT and Equations of Motion

Multi-Step Motion

Name _____

EA1

A car moving at 20 m/s brakes with constant deceleration 4 m/s² until it stops. Find the braking distance.

EA2

A particle starts at 7 m/s and accelerates uniformly at 2 m/s² over a displacement of 44 m. Find its final speed.

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KAIZEN MATHS

Exit ticket

SUVAT and Equations of Motion

Multi-Step Motion

Name _____

EB1

A cyclist starts at 5 m/s and accelerates at 4 m/s² for 4 s. The cyclist then continues at the new constant speed for 8 s. Find the total distance travelled.

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Show your working

Teacher answers & working

SUVAT and Equations of Motion · Multi-Step Motion

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S1

A car moving at 18 m/s brakes with constant deceleration 2 m/s² until it stops. Find the braking distance.

Answer:

$s = 81$ m

Working:

Choose a positive direction and list the known values.

$u = 18$ m/s, $v = 0$ m/s, $a = -2$ m/s^2, $s = ?$

Choose the equation without time.

$v^2 = u^2 + 2as$

Substitute and solve for s.

$0^2 = 18^2 + 2 \times (-2) \times s$

$0 = 324 - 4s$

$4s = 324$

$s = 324 \div 4 = 81$

This is the braking distance. Stopping distance would also include the thinking distance.

The braking distance is 81 m.

S2

A particle starts at 6 m/s and accelerates uniformly at 4 m/s² over a displacement of 176 m. Find its final speed.

Answer:

$v = 38$ m/s

Working:

List the known values.

$u = 6$ m/s, $a = 4$ m/s^2, $s = 176$ m, $v = ?$

Choose the equation without time.

$v^2 = u^2 + 2as$

Substitute the values.

$v^2 = 6^2 + 2 \times 4 \times 176$

$v^2 = 36 + 1408 = 1444$

$v = \sqrt{1444} = 38$

The final speed is 38 m/s.

Teacher answers & working

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S3

A cyclist starts at 9 m/s and accelerates at 4 m/s² for 5 s. The cyclist then continues at the new constant speed for 6 s. Find the total distance travelled.

Answer:

total distance = 269 m

Working:

Find the speed after the acceleration stage.

$v = u + at = 9 + 4 \times 5 = 29 \text{ m/s}$

Find the distance during the acceleration stage.

$s_1 = ut + \frac{1}{2}at^2$

$s_1 = 9 \times 5 + \frac{1}{2} \times 4 \times 5^2 = 95 \text{ m}$

Find the distance during the constant speed stage.

$s_2 = vt = 29 \times 6 = 174 \text{ m}$

Add the two distances.

total distance = 95 + 174 = 269

The total distance travelled is 269 m.

S4

A particle accelerates uniformly from 9 m/s to 21 m/s in 9 s. Use average velocity to find the displacement.

Answer:

$s = 135 \text{ m}$

Working:

For constant acceleration, average velocity is the mean of initial and final velocity.

average velocity = ½(u + v)

average velocity = ½(9 + 21) = 15 m/s

Multiply average velocity by time.

s = average velocity × time

$s = 15 \times 9 = 135$

The displacement is 135 m.

Teacher answers & working

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EA1

A car moving at 20 m/s brakes with constant deceleration 4 m/s² until it stops. Find the braking distance.

Answer:

$s = 50$ m

Working:

Choose a positive direction and list the known values.

$u = 20$ m/s, $v = 0$ m/s, $a = -4$ m/s², $s = ?$

Choose the equation without time.

$v^2 = u^2 + 2as$

Substitute and solve for s.

$0^2 = 20^2 + 2 \times (-4) \times s$

$0 = 400 - 8s$

$8s = 400$

$s = 400 \div 8 = 50$

This is the braking distance. Stopping distance would also include the thinking distance.

The braking distance is 50 m.

EA2

A particle starts at 7 m/s and accelerates uniformly at 2 m/s² over a displacement of 44 m. Find its final speed.

Answer:

$v = 15$ m/s

Working:

List the known values.

$u = 7$ m/s, $a = 2$ m/s², $s = 44$ m, $v = ?$

Choose the equation without time.

$v^2 = u^2 + 2as$

Substitute the values.

$v^2 = 7^2 + 2 \times 2 \times 44$

$v^2 = 49 + 176 = 225$

$v = \sqrt{225} = 15$

The final speed is 15 m/s.

Teacher answers & working

SUVAT and Equations of Motion · Multi-Step Motion

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EB1

A cyclist starts at 5 m/s and accelerates at 4 m/s² for 4 s. The cyclist then continues at the new constant speed for 8 s. Find the total distance travelled.

Answer:

total distance = 220 m

Working:

Find the speed after the acceleration stage.

$v = u + at = 5 + 4 \times 4 = 21 \text{ m/s}$

Find the distance during the acceleration stage.

$s_1 = ut + \frac{1}{2}at^2$

$s_1 = 5 \times 4 + \frac{1}{2} \times 4 \times 4^2 = 52 \text{ m}$

Find the distance during the constant speed stage.

$s_2 = vt = 21 \times 8 = 168 \text{ m}$

Add the two distances.

total distance = 52 + 168 = 220

The total distance travelled is 220 m.